

ECE276 Project3

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I. INTRODUCTION

Visual-Inertial SLAM (Simultaneous Localization and Mapping) aims to estimate the trajectory of a robot while simultaneously reconstructing the surrounding environment. This problem is fundamental in robotics, enabling applications such as autonomous navigation and mapping.

In this project, we implement an Extended Kalman Filter (EKF)-based visual-inertial SLAM system using IMU measurements and stereo visual features. The pipeline consists of IMU-based motion prediction, landmark mapping through visual observations, and full SLAM integration to refine trajectory estimation.

II. PROBLEM FORMULATION

We aim to estimate:

- Robot pose: $T_t \in SE(3)$
- Landmark positions: $m \in \mathbb{R}^{3 \times M}$

Inputs:

- IMU measurements: linear velocity v_t and angular velocity ω_t
- Stereo feature observations: z_t

Goal:

- Estimate trajectory over time
- Recover 3D landmark positions

III. TECHNICAL APPROACH

A. Part 1: IMU Localization (Prediction)

We use IMU measurements to propagate the robot pose using SE(3) kinematics:

$$T_{t+1} = T_t \cdot \exp(\hat{\xi}_t \Delta t) \quad (1)$$

where ξ_t is constructed from linear and angular velocities. The EKF prediction step updates:

$$\mu_{t+1} = f(\mu_t, u_t) \quad (2)$$

$$\Sigma_{t+1} = F_t \Sigma_t F_t^T + Q_t \quad (3)$$

B. Part 3: Landmark Mapping

We estimate landmark positions using stereo observations.

Triangulation is performed from left-right image correspondences, and EKF update is applied:

$$K_t = \Sigma_0 H^T (H \Sigma_0 H^T + V)^{-1} \quad (4)$$

$$\mu = \mu_0 + K_t(z - h(\mu_0)) \quad (5)$$

$$\Sigma = (I - K_t H) \Sigma_0 \quad (6)$$

To improve efficiency, only a subset of stable landmarks is used.

C. Part 4: Visual-Inertial SLAM

We combine IMU prediction and landmark update into a unified EKF framework.

- IMU provides motion prediction
- Visual landmarks correct drift

This significantly reduces accumulated error in trajectory estimation.

IV. RESULTS

A. Part 1: IMU Trajectory

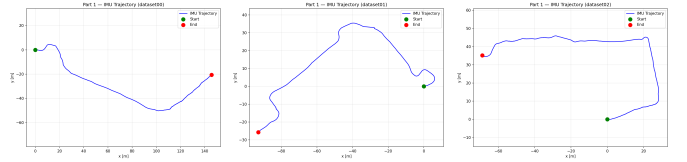


Fig. 1: IMU-only trajectory estimation.

Observation: IMU-only trajectory suffers from significant drift due to integration of noisy measurements.

B. Part 3: Landmark Mapping

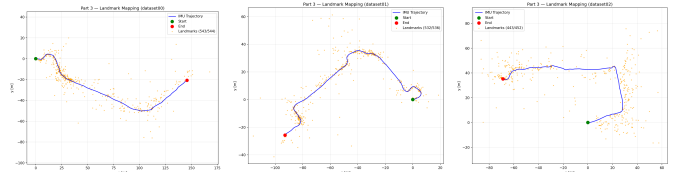


Fig. 2: Landmark mapping results.

Observation: Landmarks are distributed around the trajectory and generally align with the robot motion. Some outliers exist due to noise and imperfect feature tracking.

C. Part 4: Visual-Inertial SLAM

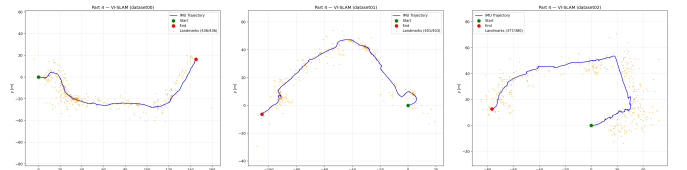


Fig. 3: Final VI-SLAM trajectory with landmarks.

Observation: Compared to Part 1, the trajectory is significantly improved. Visual updates correct drift and produce a more consistent path.

D. Video Results

In addition to the static plots, we also generated trajectory visualization videos for all three datasets:

- trajectory_video_dataset00.mp4
- trajectory_video_dataset01.mp4
- trajectory_video_dataset02.mp4

These videos provide qualitative validation of the SLAM performance and are included in the submission.

V. DISCUSSION

What works:

- EKF successfully integrates IMU and visual information
- Landmark updates reduce drift significantly

Limitations:

- Landmark depth estimation is noisy (expected due to limited motion in z-axis)
- Some outliers remain due to feature tracking errors

Future improvements:

- Better feature filtering (e.g., RANSAC)
- More robust data association
- Bundle adjustment refinement

VI. CONCLUSION

In this project, we implemented a complete visual-inertial SLAM system using EKF. The combination of IMU prediction and visual updates effectively reduces drift and improves trajectory estimation. The results demonstrate the importance of sensor fusion in robotics applications.